

AFL Prediction Project - Data Dictionary & Target Definitions

Week 1 | Data: 1983-2025 | 20 Teams | 3,109 Players | 3 Source Files

1. Source Tables

Table	Grain	Rows	Cols	Key Columns
afl_comb_CLdata	player x season	25,201	69	player_id, year, team, is_finals
players_round_by_round_clean	player x match	274,079	36	player_id, match_date, team, round
rbr_H_A_merged	player x match + venue	134,413	39	player_id, match_date, team, home_away, venue

Join keys: player_id | team + match_date (match join) | year (season aggregates)

2. Prediction Targets

Target	Definition	Formula / Rule	Level	Type
win	Team wins the match	result='W'->1, 'L'->0, draws excluded	team-match	Binary clf
margin	Winning margin (reg. alt.)	score_for - score_against (signed)	team-match	Regression
is_top_disposal	Top disposal getter on team	disposals == max(team, match)	player-match	Binary flag
is_top_goalkicker	Top goal kicker in game	goals == max(all players, match) AND goals > 0	player-match	Binary flag
afl_score	Brownlow-style composite	See formula below	player-match	Numeric

afl_score = (disposals x 1.0) + (goals x 6.0) + (marks x 1.5) + (tackles x 1.0) + (clearances x 1.5) + (inside_50s x 0.5) + (rebound_50s x 0.5) + (goal_assist x 1.0) [NaN=0]

afl_score correlates r=0.97 with official fantasy_points. Primary model target: win (binary clf).

3. Engineered Feature Dictionary (afl_feature_table_v1.csv)

Feature	Type	Window	Source	Description
roll3_win / roll5_win	numeric	last 3/5 games	result	Team win rate over last N matches (shift(1) no leakage)
roll3/5_goals, disposals, fp	numeric	last 3/5 games	stats	Team totals for goals/disposals/fantasy points over last N matches
win_streak / loss_streak	integer	current	result	Consecutive W or L streak going into match (capped at 5)
h2h_winrate / h2h_cumgames	numeric	all history	result+opp	Historical win rate and games played vs this specific opponent
is_home	binary	current	home_away	1 = playing at home ground, 0 = away
venue_winrate	numeric	all history	venue+result	Historical win rate at this specific venue
crowd	numeric	current	crowd	Match attendance
rest_days / rest_advantage	integer	current	match_date	Days since last match; and difference from opponent's rest
season_winpct	numeric	season to date	result	Win % this season before this match (ladder position proxy)
season_cumwins / cumgames	integer	season to date	result	Season record (W and games) accumulated before this match

4. Train / Hold-Out Split

Train: 1983-2023 (13,790 team-match rows). **Hold-out:** 2024 (396 rows). **Split rationale:** Time-based only - random splits leak future rolling features into training labels. **Realistic accuracy ceiling:** ~65-68% binary W/L. Models >75% on unseen future seasons = leakage red flag.

5. Key EDA Findings

- Home advantage: home teams win 59.0% vs 40.3% away (2000-2025). COVID hub 2020 eliminated it.
- Hawthorn and Geelong lead all-time win rates (>58%); Gold Coast and GWS below 40% (expansion teams).
- Rest days: 8-9 day rest wins at 51% vs 49% for 6-7 days - modest but consistent effect.
- Missing data is era-driven: clearances, inside 50s not recorded pre-1994. Use disposals/kicks/tackles for full-history models.
- Margin distribution approximately symmetric (std ~44pts). Draws are 0.86% of matches.
- Player fantasy points CV averages 0.38 (std/mean) - high week-to-week noise. Form windows are essential.